

ILLUSTRATIONS

Illustration-01: Find first derivative of the following functions:

(a) $7x^4 + 3x^3 - 9x + 5$, (b) $x + \frac{4}{x} - \frac{2}{x^2}$.

Solution: (a) Let $y = 7x^4 + 3x^3 - 9x + 5$

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(7x^4 + 3x^3 - 9x + 5)$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}(7x^4) + \frac{d}{dx}(3x^3) - \frac{d}{dx}(9x) + \frac{d}{dx}(5)$$

$$\Rightarrow \frac{dy}{dx} = 4 \cdot 7x^{4-1} + 3 \cdot 3x^{3-1} - 9 \cdot 1x^{1-1} + 0 \Rightarrow \frac{dy}{dx} = 28x^3 + 9x^2 - 9 \text{ [Ans.]}$$

(b) Let $y = x + \frac{4}{x} - \frac{2}{x^2} \Rightarrow y = x + 4x^{-1} - 2x^{-2}$

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(x + 4x^{-1} - 2x^{-2}) \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(x) + \frac{dy}{dx}(4x^{-1}) - \frac{dy}{dx}(2x^{-2})$$

$$\Rightarrow \frac{dy}{dx} = 1 + 4(-1)x^{-1-1} - 2(-2)x^{-2-1}$$

$$\Rightarrow \frac{dy}{dx} = 1 - 4x^{-2} + 4x^{-3} \Rightarrow \frac{dy}{dx} = 1 - \frac{4}{x^2} + \frac{4}{x^3} \text{ [Ans.]}$$

Illustration-02: Find first derivative of the following functions:

$$(a) \frac{3x^2 + 5x}{7x + 4}, \quad (b) \frac{(2x+1)(3x+1)}{4x+1}$$

Solution: (a) Let $y = \frac{3x^2 + 5x}{7x + 4} \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx} \left(\frac{3x^2 + 5x}{7x + 4} \right)$

$$\Rightarrow \frac{dy}{dx} = \frac{(7x+4) \frac{d}{dx}(3x^2 + 5x) - (3x^2 + 5x) \frac{d}{dx}(7x+4)}{(7x+4)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(7x+4)(6x+5) - (3x^2 + 5x)(7)}{(7x+4)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{42x^2 + 35x + 24x + 20 - 21x^2 - 35x}{(7x+4)^2} \Rightarrow \frac{dy}{dx} = \frac{21x^2 + 24x + 20}{(7x+4)^2} \text{ [Ans.]}$$

(b) Let $y = \frac{(2x+1)(3x+1)}{4x+1} \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx} \frac{6x^2 + 5x + 1}{4x+1}$

$$\Rightarrow \frac{dy}{dx} = \frac{(4x+1) \frac{d}{dx}(6x^2 + 5x + 1) - (6x^2 + 5x + 1) \frac{d}{dx}(4x+1)}{(4x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(4x+1)(12x+5) - (6x^2 + 5x + 1)(4)}{(4x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{48x^2 + 20x + 12x + 5 - 24x^2 - 20x - 4}{(4x+1)^2} \Rightarrow \frac{dy}{dx} = \frac{24x^2 + 12x + 1}{(4x+1)^2} \text{ [Ans.]}$$

Illustration-03: Find first derivative of the following functions:

$$(a) (5x^3 + 6x^2 + 11x + 7)^{11}, \quad (b) \frac{1}{\sqrt{2x^4 + 3x^3 - 5x + 6}}$$

Solution: (a) Let $y = (5x^3 + 6x^2 + 11x + 7)^{11}$

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx} (5x^3 + 6x^2 + 11x + 7)^{11}$$

$$\Rightarrow \frac{dy}{dx} = 11(5x^3 + 6x^2 + 11x + 7)^{11-1} \frac{d}{dx}(5x^3 + 6x^2 + 11x + 7)$$

$$\Rightarrow \frac{dy}{dx} = 11(5x^3 + 6x^2 + 11x + 7)^{10} (15x^2 + 12x + 11) \text{ [Ans.]}$$

$$\begin{aligned}
 \text{(b) Let } y &= \frac{1}{\sqrt[3]{2x^4 + 3x^3 - 5x + 6}} \\
 \Rightarrow \frac{dy}{dx} &= \frac{d}{dx} (2x^4 + 3x^3 - 5x + 6)^{-\frac{1}{3}} \\
 \Rightarrow \frac{dy}{dx} &= \frac{1}{3} (2x^4 + 3x^3 - 5x + 6)^{-\frac{1}{3}-1} \frac{d}{dx} (2x^4 + 3x^3 - 5x + 6) \\
 \Rightarrow \frac{dy}{dx} &= -\frac{1}{3} (2x^4 + 3x^3 - 5x + 6)^{-\frac{4}{3}} (8x^3 + 9x^2 - 5) \\
 \Rightarrow \frac{dy}{dx} &= -\frac{1}{3} \cdot \frac{1}{(2x^4 + 3x^3 - 5x + 6)^{\frac{4}{3}}} \cdot (8x^3 + 9x^2 - 5) \\
 \Rightarrow \frac{dy}{dx} &= -\frac{(8x^3 + 9x^2 - 5)}{3 \cdot \sqrt[3]{(2x^4 + 3x^3 - 5x + 6)^4}} \quad [\text{Ans.}]
 \end{aligned}$$

Illustration-04: Find first derivative of the following functions:

(a) $3\sqrt{x} \sin x$, (b) $x^3 \tan x$.

Solution: (a) Let $y = 3\sqrt{x} \sin x \Rightarrow \frac{d}{dx}(y) = 3 \frac{d}{dx} [\sqrt{x} \sin x]$

$$\begin{aligned}
 \Rightarrow \frac{dy}{dx} &= 3 \left[\sqrt{x} \frac{d}{dx} (\sin x) + \sin x \frac{d}{dx} (\sqrt{x}) \right] \Rightarrow \frac{dy}{dx} = 3 \left[\sqrt{x} \cos x + \sin x \cdot \frac{1}{2\sqrt{x}} \right] \\
 \Rightarrow \frac{dy}{dx} &= 3 \left[\frac{2x \cos x}{2\sqrt{x}} + \frac{\sin x}{2\sqrt{x}} \right] \Rightarrow \frac{dy}{dx} = \frac{3}{2\sqrt{x}} (2x \cos x + \sin x) \quad [\text{Ans.}]
 \end{aligned}$$

(b) Let $y = x^3 \tan x \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx} [x^3 \tan x]$

$$\begin{aligned}
 \Rightarrow \frac{dy}{dx} &= x^3 \frac{d}{dx} (\tan x) + \tan x \frac{d}{dx} (x^3) \Rightarrow \frac{dy}{dx} = x^3 \cdot \sec^2 x + \tan x \cdot 3x^2 \\
 \Rightarrow \frac{dy}{dx} &= 3x^2 \tan x + x^3 \sec^2 x \Rightarrow \frac{dy}{dx} = x^2 (3 \tan x + x \sec^2 x) \quad [\text{Ans.}]
 \end{aligned}$$

Illustration-05: (i) If $y = e^x \cos x$, (ii) $y = x^3 \log x$, then find $\frac{dy}{dx}$.

Solution:

$$\frac{dy}{dx} = \frac{d}{dx} (e^x \cos x) = e^x \frac{d}{dx} (\cos x) + \cos x \frac{d}{dx} (e^x) = -e^x \sin x + \cos x e^x = e^x (\cos x - \sin x).$$

(ii) If $y = x^3 \log x$, then find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx}(x^3 \log x) = x^3 \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x^3) = x^3 \times \frac{1}{x} + \log x \times 3x^2 = x^2(1 + 3 \log x)$$

Illustration-06: (i) If $y = \frac{e^x}{\log x}$, then

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{e^x}{\log x}\right) = \frac{\log x \frac{d}{dx}(e^x) - e^x \frac{d}{dx}(\log x)}{(\log x)^2} = \frac{\log x \cdot e^x - e^x(1/x)}{(\log x)^2}$$

(ii) If $y = \frac{4}{x^3}$, then

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{4}{x^3}\right) = \frac{x^3 \frac{d}{dx}(4) - 4 \frac{d}{dx}(x^3)}{(x^3)^2} = \frac{x^3(0) - 4(3x^2)}{x^6} = \frac{-12x^2}{x^6} = \frac{-12}{x^4}$$

Illustration-07: (a) If $y = \sqrt{\frac{1-x}{1+x}}$ prove that $(1-x)^2 \frac{dy}{dx} + y = 0$

(b) If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, show that $\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$

Solution: (a) $y = \sqrt{\frac{1-x}{1+x}} = \sqrt{\frac{1-x}{1+x} \times \frac{1-x}{1-x}} = \frac{1-x}{\sqrt{1-x^2}}$

$$\therefore \frac{dy}{dx} = \frac{\sqrt{1-x^2}(0-1) - (1-x)\left\{\frac{1}{2}(1-x^2)^{-1/2}(-2x)\right\}}{(\sqrt{1-x^2})^2} = \frac{-\sqrt{1-x^2} + x(1-x)(1-x^2)^{-1/2}}{1-x^2}$$

$$= \frac{-\sqrt{1-x^2} + x \frac{(1-x)}{\sqrt{1-x^2}}}{1-x^2}$$

$$\text{or, } (1-x^2) \frac{dy}{dx} = -\sqrt{1-x^2} + \frac{x(1-x)}{\sqrt{1-x^2}} = \frac{-(1-x^2) + x(1-x)}{\sqrt{1-x^2}} = \frac{-1+x^2+x-x^2}{\sqrt{1-x^2}} = \frac{-(1-x)}{\sqrt{1-x^2}} = -y$$

$$\text{Hence, } (1-x^2) \frac{dy}{dx} + y = 0.$$

$$\text{(b) } y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}} = \left[\frac{\cos^2 x + \sin^2 x - 2 \sin x \cos x}{\cos^2 x + \sin^2 x + 2 \sin x \cos x} \right]^{1/2} = \left[\frac{(\cos x - \sin x)^2}{(\cos x + \sin x)^2} \right]^{1/2} = \frac{\cos x - \sin x}{\cos x + \sin x} = \frac{1 - \tan x}{1 + \tan x}$$

on dividing by $\cos x = \frac{\tan(\pi/4) - \tan x}{\tan(\pi/4) + \tan x} = \tan\left(\frac{\pi}{4} - x\right)$

$$\therefore \frac{dy}{dx} = \sec^2\left(\frac{\pi}{4} - x\right) \frac{d}{dx}\left(\frac{\pi}{4} - x\right) = -\sec^2\left(\frac{\pi}{4} - x\right) \text{ or } \frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0.$$

Illustration-08: Find $\frac{dy}{dx}$ where $y = (x^2 + 3)^3$.

Solution: Given $y = (x^2 + 3)^3$.

Therefore, $\frac{dy}{dx} = 3(x^2 + 3)^2 \frac{d}{dx}(x^2 + 3) = 3(x^2 + 3)^2 (2x) = 6x(x^2 + 3)^2$

Illustration-09: If $\sin y = x \sin(a + y)$, show that $\frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a}$

Solution: Given, $\sin y = x \sin(a + y) \Rightarrow x = \frac{\sin y}{\sin(a + y)}$

$$\therefore \frac{dx}{dy} = \frac{\sin(a + y) \cos y - \sin y \cos(a + y)}{\sin^2(a + y)} = \frac{\sin(a + y - y)}{\sin^2(a + y)}$$

$$[\because \sin(A - B) = \sin A \cos B - \cos A \sin B]$$

$$= \frac{\sin a}{\sin^2(a + y)} \therefore \frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a} \text{ (proved)}$$

Illustration-10: If $y = x^3 \log\left(\frac{1}{x}\right)$ prove that $x \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 3x^2 = 0$

Solution: We have $y = x^3 \log\left(\frac{1}{x}\right)$

Differentiating with respect to x by parts, we get

$$\frac{dy}{dx} = x^3 \frac{d}{dx} \left[\log\left(\frac{1}{x}\right) \right] + \log\left(\frac{1}{x}\right) \frac{d}{dx}(x^3)$$

$$= x^3(x) \left(-\frac{1}{x^2} \right) + \log\left(\frac{1}{x}\right) (3x^2) = -x^2 + 3x^2 \log\left(\frac{1}{x}\right)$$

$$\therefore x \frac{dy}{dx} = -x^3 + 3x^3 \log\left(\frac{1}{x}\right) = -x^3 + 3y$$

Differentiating both sides again, we get

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = -3x^2 + 3 \frac{dy}{dx}$$

$$\text{or, } x \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 3x^2 = 0$$

Illustration-11: If $y = \log \left\{ x + \sqrt{x^2 + a^2} \right\}$ prove that $(a^2 + x^2) \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} = 0$

Solution: Given $y = \log \left\{ x + \sqrt{x^2 + a^2} \right\}$

Differentiating with respect to x , we have

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + a^2}} \left\{ 1 + \frac{1}{2} \frac{1}{\sqrt{x^2 + a^2}} \cdot 2x \right\} = \frac{1}{\sqrt{x^2 + a^2}}$$

Squaring and cross multiplying, we get, $(x^2 + a^2) \left(\frac{dy}{dx} \right)^2 = 1$

Differentiating again, we have

$$(a^2 + x^2) 2 \left(\frac{dy}{dx} \right) \frac{d^2 y}{dx^2} + 2x \left(\frac{dy}{dx} \right)^2 = 0$$

Dividing both sides by $2 \frac{dy}{dx}$, we get

$$(a^2 + x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0$$

Illustration-12: Differentiate x^x with respect to x .

Solution: Let $y = x^x \Rightarrow \log y = \log x^x$ [Taking 'log' on both sides]

$$\Rightarrow \log y = x \log x \Rightarrow \frac{d}{dx} (\log y) = x \frac{d}{dx} (\log x) + \log x \frac{d}{dx} (x)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{d}{dx} (y) = x \cdot \frac{1}{x} + \log x \cdot 1 \Rightarrow \frac{dy}{dx} = y(1 + \log x) \Rightarrow \frac{dy}{dx} = x^x (1 + \log x) \text{ [Ans.]}$$

Illustration-13: Differentiate the following functions w.r.t. x :

(a) $(1+x)^{2x}$, (b) $x^{1/x}$, (c) $(x^x)^x$.

Solution: (a) Let $y = (1+x)^{2x}$

$$\Rightarrow \log y = \log(1+x)^{2x} \text{ [Taking 'log' on both sides]}$$

$$\Rightarrow \log y = 2x \log(1+x) \Rightarrow \frac{d}{dx}(\log y) = 2x \frac{d}{dx}[\log(1+x)] + \log(1+x) \frac{d}{dx}(2x)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = 2x \cdot \frac{1}{1+x} (0+1) + \log(1+x) \cdot 2$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{2x}{1+x} + 2 \log(1+x) \right] \Rightarrow \frac{dy}{dx} = 2y \left[\frac{x}{x+1} + \log(x+1) \right]$$

$$\Rightarrow \frac{dy}{dx} = 2(1+x)^{2x} \left[\frac{x}{x+1} + \log(x+1) \right] \text{ [Ans.]}$$

(b) Let $y = x^{1/x}$

$$\Rightarrow \log y = \log x^{1/x} \text{ [Taking 'log' on both sides]} \Rightarrow \log y = \frac{1}{x} \log x$$

$$\Rightarrow \frac{d}{dx}(\log y) = \frac{1}{x} \frac{d}{dx}(\log x) + \log x \frac{d}{dx} \left(\frac{1}{x} \right) \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} \cdot \frac{1}{x} + \log x \left(-\frac{1}{x^2} \right)$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{1}{x^2} - \frac{\log x}{x^2} \right] \Rightarrow \frac{dy}{dx} = x^{1/x} \left[\frac{1 - \log x}{x^2} \right] \text{ [Ans.]}$$

(c) Let $y = (x^x)^x \Rightarrow \log y = \log x^{x^2}$ [Taking 'log' on both sides] $\Rightarrow \log y = x^2 \log x$

$$\Rightarrow \frac{d}{dx}(\log y) = x^2 \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x^2) \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = x^2 \cdot \frac{1}{x} + \log x (2x)$$

$$\Rightarrow \frac{dy}{dx} = y[x + 2x \log x] \Rightarrow \frac{dy}{dx} = yx[1 + 2 \log x] \Rightarrow \frac{dy}{dx} = (x^x)^x \cdot x[1 + 2 \log x]$$

$$\Rightarrow \frac{dy}{dx} = x^{x^2} \cdot x^x [1 + 2 \log x] \Rightarrow \frac{dy}{dx} = x^{x^2+x} (1 + 2 \log x) \text{ [Ans.]}$$

Illustration-14: If $y = x^{x^x}$, find $\frac{dy}{dx}$.

Solution: We have $y = x^{x^x}$. Taking logarithm on both sides, we get $\log y = x^x \log x$
Differentiating both sides with respect to x

$$\frac{1}{y} \frac{dy}{dx} = x^x \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x^x)$$

$$= x^x (1/x) + \log x [x^x (1 + \log x)]$$

$$\therefore \frac{dy}{dx} = x^{x^x} [x^{x-1} + \log x \cdot x^x (1 + \log x)]$$

$$\left[\frac{d}{dx}(u^v) = u^v \left[v \frac{d}{dx} \ln u + \ln u \frac{dv}{dx} \right] \right]$$

Illustration-15: If $y = 3^x e^{2x} \log(x^3 + 1)$, find $\frac{dy}{dx}$.

Solution: We have $y = 3^x e^{2x} \log(x^3 + 1)$. Taking logarithm on both sides, we get
 $\log y = \log(3^x) + \log(e^{2x}) + \log[\log(x^3 + 1)]$

Differentiating both sides with respect to x we have

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{3^x} \frac{d}{dx}(3^x) + \frac{1}{e^{2x}} \frac{d}{dx}(e^{2x}) + \frac{1}{\log(x^3 + 1)} \frac{d}{dx}[\log(x^3 + 1)]$$

$$= \frac{1}{3^x} 3^x \log 3 + \frac{1}{e^{2x}} \times 2(e^{2x}) + \frac{1}{\log(x^3 + 1)} \times \frac{1}{x^3 + 1} \times (3x^2)$$

$$= \log 3 + 2 + \frac{1}{\log(x^3 + 1)} \times \frac{3x^2}{x^3 + 1}$$

$$\therefore \frac{dy}{dx} = y \left[\log 3 + 2 + \frac{1}{\log(x^3 + 1)} \times \frac{3x^2}{x^3 + 1} \right]$$

$$\therefore \frac{dy}{dx} = 3^x e^{2x} \log(x^3 + 1) \left[\log 3 + 2 + \frac{1}{\log(x^3 + 1)} \times \frac{3x^2}{x^3 + 1} \right]$$

$$\therefore \frac{dy}{dx} = 3^x e^{2x} \log(x^3 + 1) \left[\log 3 + 2 + \frac{3x^2}{(x^3 + 1) \log(x^3 + 1)} \right]$$

Illustration-16: Find $\frac{dy}{dx}$ from the equation $\log(y/x) = \sec x$.

Solution: Given $\log\left(\frac{y}{x}\right) = \log y - \log x = \sec x$

Differentiating both sides w. r. to x , we get

$$\frac{1}{y} \frac{dy}{dx} - \frac{1}{x} = \sec x \tan x$$

$$\text{or, } \frac{1}{y} \frac{dy}{dx} = \frac{1}{x} + \sec x \tan x$$

$$\text{or, } \frac{dy}{dx} = y \left(\frac{1}{x} + \sec x \tan x \right) = \frac{y}{x} (1 + x \sec x \tan x)$$

$$= e^{\sec x} (1 + x \sec x \tan x) \quad \left[\because \log(y/x) = \sec x \Rightarrow y/x = e^{\sec x} \right]$$

Illustration-20: Find first derivative of the following functions:

(a) $\frac{\sin 2x}{1 + \cos 2x}$, (b) $\frac{\cos x}{\cos x + \sin x}$.

Solution:

(a) Let $\frac{\sin 2x}{1 + \cos 2x}$,

$$\Rightarrow y = \frac{\sin x}{\cos x} \Rightarrow y = \tan x \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(\tan x) \Rightarrow \frac{dy}{dx} = \sec^2 x \text{ [Ans.]}$$

(b) Let $\frac{\cos x}{\cos x + \sin x} \Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left[\frac{\cos x}{\cos x + \sin x} \right]$

$$\Rightarrow \frac{dy}{dx} = \frac{(\cos x + \sin x) \frac{dy}{dx}(\cos x) - \cos x \frac{dy}{dx}(\cos x + \sin x)}{(\cos x + \sin x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(\cos x + \sin x)(-\sin x) - \cos x(-\sin x + \cos x)}{(\cos x + \sin x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\sin x \cos x - \sin^2 x + \sin x \cos x - \cos^2 x}{(\cos x + \sin x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\sin^2 x - \cos^2 x}{(\cos x + \sin x)^2} \Rightarrow \frac{dy}{dx} = \frac{-(\sin^2 x + \cos^2 x)}{(\cos x + \sin x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{(\cos x + \sin x)^2} \text{ [Ans.]} \quad [\because \sin^2 x + \cos^2 x = 1]$$

Illustration-21: If $y = \sin(2 \sin^{-1} x)$, show that $\frac{dy}{dx} = 2 \sqrt{1 - x^2}$.

Solution: Given $y = \sin(2 \sin^{-1} x) \Rightarrow \sin^{-1} y = 2 \sin^{-1} x$

$$\Rightarrow \frac{d}{dx}(\sin^{-1} y) = \frac{d}{dx}(2 \sin^{-1} x) \Rightarrow \frac{1}{\sqrt{1 - y^2}} \frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1 - x^2}}$$

$$\Rightarrow \frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1-x^2}} \times \frac{\sqrt{1-y^2}}{1} \Rightarrow \frac{dy}{dx} = 2 \cdot \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} \Rightarrow \frac{dy}{dx} = 2\sqrt{\frac{1-y^2}{1-x^2}} \quad [\text{Shown}]$$

Illustration-22: Find first derivative of the following functions:

(a) $\log \sin x$, (b) $e^{\log \sin x}$

Solution: (a) Let $y = \log \sin x \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(\log \sin x) \Rightarrow \frac{dy}{dx} = \frac{1}{\sin x} \cdot \cos x$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x}{\sin x} \Rightarrow \frac{dy}{dx} = \cot x \quad [\text{Ans.}]$$

(b) Let $y = e^{\log \sin x}$

$$\Rightarrow \log y = \log e^{\log \sin x} \quad [\text{Taking 'log' on both sides}]$$

$$\Rightarrow \log y = \log \sin x \cdot \log e \Rightarrow \log y = \log \sin x \cdot 1 \quad [\because \log e = 1]$$

$$\Rightarrow \log y = \log \sin x \Rightarrow y = \sin x \quad [\text{Omitting 'log' on both sides}]$$

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(\sin x) \Rightarrow \frac{dy}{dx} = \cos x \quad [\text{Ans.}]$$

Illustration-23: Differentiate $\sin(\log x)$ w.r.t. $\tan(e^x)$

Solution: Let $y = \sin(\log x) \Rightarrow \frac{dy}{dx} = \frac{d}{dx}[\sin(\log x)]$

$$\Rightarrow \frac{dy}{dx} = \cos(\log x) \cdot \frac{1}{x} \Rightarrow \frac{dy}{dx} = \frac{\cos(\log x)}{x} \Rightarrow \frac{dy}{dx} = \cos(\log x) / x$$

Also, let $z = \tan(e^x) \Rightarrow \frac{dz}{dx} = \frac{d}{dx}[\tan(e^x)] \Rightarrow \frac{dz}{dx} = (\sec^2 e^x) e^x$

$$\Rightarrow \frac{dz}{dx} = e^x (\sec^2 e^x) \therefore \frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} \Rightarrow \frac{dy}{dz} = \frac{\cos(\log x) / x}{e^x (\sec^2 e^x)} \quad [\text{Ans.}]$$

Illustration-24: Find $\frac{dy}{dx}$ in the case of $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

Solution: Given $x = a(t - \sin t)$

$$\Rightarrow \frac{dx}{dt} = \frac{d}{dt}[a(t - \sin t)] \Rightarrow \frac{dx}{dt} = a \left[\frac{d}{dt}(t) - \frac{d}{dt}(\sin t) \right]$$

$$\Rightarrow \frac{dx}{dt} = a(1 - \cos t) \Rightarrow \frac{dx}{dt} = a \cdot 2 \sin^2 \frac{t}{2} \left[\because 1 - \cos t = 2 \sin^2 \frac{t}{2} \right]$$

Also, given $y = a(1 - \cos t) \Rightarrow \frac{dx}{dt} = a2 \sin^2 \frac{t}{2} \left[\because 1 - \cos t = 2 \sin^2 \frac{t}{2} \right]$

$$\Rightarrow \frac{dy}{dt} = \frac{dx}{dt} \left[2a \sin^2 \frac{t}{2} \right] \Rightarrow \frac{dy}{dt} = 2a \left(2 \sin \frac{t}{2} \right) \left(\cos \frac{t}{2} \right) \left(\frac{1}{2} \right)$$

$$\Rightarrow \frac{dy}{dt} = 2a \sin \frac{t}{2} \cos \frac{t}{2}$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \Rightarrow \frac{dy}{dx} = \frac{2a \sin \frac{t}{2} \cos \frac{t}{2}}{a \cdot 2 \sin^2 \frac{t}{2}} \Rightarrow \frac{dy}{dx} = \frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} \Rightarrow \frac{dy}{dx} = \cot \frac{t}{2} \quad [\text{Ans.}]$$

Illustration-25: Differentiate with respect to x of the following:

~~F-12~~ (a) $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$, (b) $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$ ~~F-12~~

Solution: (a) Let $y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \Rightarrow y = \cos^{-1} \left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta} \right)$ [Putting $x = \tan \theta$]

$$\Rightarrow y = \cos^{-1}(\cos 2\theta) \left[\because \cos 2\theta = \frac{1-\tan^2 \theta}{1+\tan^2 \theta} \right]$$

$$\Rightarrow y = 2\theta \Rightarrow y = 2 \tan^{-1} x \left[\because x = \tan \theta \Rightarrow \theta = \tan^{-1} x \right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} (2 \tan^{-1} x) \Rightarrow \frac{dy}{dx} = 2 \cdot \frac{1}{1+x^2} \Rightarrow \frac{dy}{dx} = \frac{2}{1+x^2} \quad [\text{Ans.}]$$

(b) Let $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right) \Rightarrow y = \tan^{-1} \left(\frac{2 \tan \theta}{1-\tan^2 \theta} \right)$ [Putting $x = \tan \theta$]

$$\Rightarrow y = \tan^{-1}(\tan 2\theta) \left[\because \tan 2\theta = \frac{2 \tan \theta}{1-\tan^2 \theta} \right]$$

$$\Rightarrow y = 2\theta \Rightarrow y = 2 \tan^{-1} x \left[\because x = \tan \theta \Rightarrow \theta = \tan^{-1} x \right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} (2 \tan^{-1} x) \Rightarrow \frac{dy}{dx} = 2 \cdot \frac{1}{1+x^2} \Rightarrow \frac{dy}{dx} = \frac{2}{1+x^2} \quad [\text{Ans.}]$$

Illustration-26: Differentiate with respect to x of the following:

(a) $\tan^{-1} \left(\frac{\sin x}{1+\cos x} \right)$, (b) $\tan^{-1} \left(\sqrt{\frac{1-\cos x}{1+\cos x}} \right)$

Solution: (a) Let $y = \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right)$

$$\Rightarrow y = \tan^{-1} \left(\frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right) \left[\because \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}, 1 + \cos x = 2 \cos^2 \frac{x}{2} \right]$$

$$\Rightarrow y = \tan^{-1} \left(\tan \frac{x}{2} \right) \Rightarrow y = \frac{x}{2} \Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left(\frac{x}{2} \right) \Rightarrow \frac{dy}{dx} = \frac{1}{2} \text{ [Ans.]}$$

(b) Let $y = \tan^{-1} \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right) \Rightarrow y = \tan^{-1} \left(\sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} \right) \text{ [Putting } x = 2\theta \text{]}$

$$\Rightarrow y = \tan^{-1} \left(\sqrt{\tan^2 \frac{x}{2}} \right) \Rightarrow y = \tan^{-1} \left(\tan \frac{x}{2} \right) \Rightarrow y = \frac{x}{2} \Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left(\frac{x}{2} \right) \Rightarrow \frac{dy}{dx} = \frac{1}{2} \text{ [Ans.]}$$

Illustration-32: Find $\frac{dy}{dx}$ of the following: (a) $x^2 + y^2 - 2x = 0$ (b) $x^2 + 3xy + y^2 = 4$.

Solution: (a) Given $x^2 + y^2 - 2x = 0 \Rightarrow \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(2x)$

$$\Rightarrow \frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(2x) \Rightarrow 2x + 2y \frac{d}{dx}(y) = 2$$

$$\Rightarrow 2y \frac{dy}{dx} = 2 - 2x \Rightarrow \frac{dy}{dx} = \frac{2(1-x)}{2y} \Rightarrow \frac{dy}{dx} = \frac{1-x}{y} \text{ [Ans.]}$$

(b) Given $x^2 + 3xy + y^2 = 4 \Rightarrow \frac{d}{dx}(x^2 + 3xy + y^2) = \frac{d}{dx}(4)$

$$\Rightarrow \frac{d}{dx}(x^2) + 3x \frac{d}{dx}(y) + y \frac{d}{dx}(3x) + \frac{d}{dx}(y^2) = \frac{d}{dx}(4)$$

$$\Rightarrow 2x + 3x \frac{dy}{dx} + y \cdot 3 + 2y \frac{d}{dx}(y) = 0 \Rightarrow 3x \frac{dy}{dx} + 2y \frac{dy}{dx} = -(2x + 3y)$$

$$\Rightarrow (3x + 2y) \frac{dy}{dx} = -(2x + 3y) \Rightarrow \frac{dy}{dx} = -\frac{2x + 3y}{3x + 2y} \text{ [Ans.]}$$

Illustration-33: Differentiate the following functions w.r.t.x: $\sin y = x \sin(a + y)$

Solution: Given $\sin y = x \sin(a + y) \Rightarrow x = \frac{\sin y}{\sin(a + y)} \Rightarrow \frac{d}{dy}(x) = \frac{d}{dy} \left[\frac{\sin y}{\sin(a + y)} \right]$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a + y) \frac{d}{dy}(\sin y) - \sin y \frac{d}{dy}[\sin(a + y)]}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a + y) \cos y - \sin y \cos(a + y)(0 + 1)}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a + y) \cos y - \sin y \cos(a + y)}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a + y - y)}{\sin^2(a + y)} \left[\because \sin(A - B) = \sin A \cos B - \cos A \sin B \right]$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin a}{\sin^2(a + y)} \Rightarrow \frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a} \quad [\text{Ans.}]$$

Illustration-36: (a) If $y = ax^3 + bx^2 + cx + d$, find y_1 and y_4 .

(b) If $y = \frac{\log x}{x}$, show that $\frac{d^2 y}{dx^2} = \frac{2 \log x - 3}{x^3}$.

Solution: (a) Given $y = ax^3 + bx^2 + cx + d$

$$\Rightarrow y_1 = \frac{d}{dx}(ax^3) + \frac{d}{dx}(bx^2) + \frac{d}{dx}(cx) + \frac{d}{dx}(d)$$

$$\Rightarrow y_1 = 3ax^2 + 2bx + c \Rightarrow y_2 = \frac{d}{dx}(3ax^2) + \frac{d}{dx}(2bx) + \frac{d}{dx}(c)$$

$$\Rightarrow y_2 = 6ax + 2b \Rightarrow y_3 = \frac{d}{dx}(6ax) + \frac{d}{dx}(2b) \Rightarrow y_3 = 6a \text{ [Ans.]}$$

$$\text{and } y_3 = 6a \Rightarrow y_4 = \frac{d}{dx}(6a) \Rightarrow y_4 = 0 \text{ [Ans.]}$$

$$\text{(b) Given } y = \frac{\log x}{x} \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}\left[\frac{\log x}{x}\right] \Rightarrow \frac{dy}{dx} = \frac{x \frac{d}{dx}(\log x) - \log x \frac{d}{dx}(x)}{(x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \log x(1)}{x^2} \Rightarrow \frac{dy}{dx} = \frac{1 - \log x}{x^2} \Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left[\frac{1 - \log x}{x^2}\right]$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{x^2 \frac{d}{dx}(1 - \log x) - (1 - \log x) \frac{d}{dx}(x^2)}{(x^2)^2} \Rightarrow \frac{d^2 y}{dx^2} = \frac{x^2\left(-\frac{1}{x}\right) - (1 - \log x)(2x)}{x^4}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{-x - 2x(1 - \log x)}{x^4} \Rightarrow \frac{d^2 y}{dx^2} = \frac{-x - 2x + 2x \log x}{x^4} \Rightarrow \frac{d^2 y}{dx^2} = \frac{2x \log x - 3x}{x^4}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{x(2 \log x - 3)}{x^4} \Rightarrow \frac{d^2 y}{dx^2} = \frac{2 \log x - 3}{x^3} \text{ [Shown]}$$

F-18

Illustration-40: If $y = e^x \cos 2x$, then show that $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 5y = 0$.

Solution: Given $y = e^x \cos 2x \Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}[e^x \cos 2x]$

$$\Rightarrow \frac{dy}{dx} = e^x \frac{d}{dx}(\cos 2x) + \cos 2x \frac{d}{dx}(e^x) \Rightarrow \frac{dy}{dx} = e^x(-\sin 2x)2 + \cos 2x \cdot e^x$$

$$\Rightarrow \frac{dy}{dx} = -2e^x \sin 2x + e^x \cos 2x \Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}[-2e^x \sin 2x] + \frac{d}{dx}[e^x \cos 2x]$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -2\left[e^x \frac{d}{dx}(\sin 2x) + \sin 2x \frac{d}{dx}(e^x)\right] + e^x \frac{d}{dx}(\cos 2x) + \cos 2x \frac{d}{dx}(e^x)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -2[e^x(\cos 2x)2 + \sin 2x(e^x)] + e^x(-\sin 2x)2 + \cos 2x(e^x)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -4e^x \cos 2x - 2e^x \sin 2x - 2e^x \sin 2x + e^x \cos 2x$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -3e^x \cos 2x - 4e^x \sin 2x$$

$$\therefore LHS = \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 5y$$

$$= -3e^x \cos 2x - 4e^x \sin 2x - 2(-2e^x \sin 2x + e^x \cos 2x) + 5e^x \cos 2x$$

$$= -3e^x \cos 2x - 4e^x \sin 2x + 4e^x \sin 2x - 2e^x \cos 2x + 5e^x \cos 2x$$

$$= -5e^x \cos 2x + 5e^x \cos 2x = 0 = RHS \text{ [Proved]}$$

F-18

Illustration-41: If $x = \frac{1-t}{1+t}$ and $y = \frac{2t}{1+t}$, find $\frac{d^2 y}{dx^2}$.

Solution: Given $x = \frac{1-t}{1+t} \Rightarrow \frac{d}{dt}(x) = \frac{d}{dt} \left[\frac{1-t}{1+t} \right]$

$$\Rightarrow \frac{dx}{dt} = \frac{(1+t) \frac{d}{dt}(1-t) - (1-t) \frac{d}{dt}(1+t)}{(1+t)^2} \Rightarrow \frac{dx}{dt} = \frac{(1+t)(-1) - (1-t)(1)}{(1+t)^2}$$

$$\Rightarrow \frac{dx}{dt} = \frac{-1-t-1-t}{(1+t)^2} \Rightarrow \frac{dx}{dt} = \frac{-2-2t}{(1+t)^2} \Rightarrow \frac{dx}{dt} = \frac{-2(1+t)}{(1+t)^2} \Rightarrow \frac{dx}{dt} = \frac{-2}{(1+t)}$$

Also $y = \frac{2t}{1+t} \Rightarrow \frac{d}{dt}(y) = \frac{d}{dt} \left[\frac{2t}{1+t} \right]$

$$\Rightarrow \frac{dy}{dt} = \frac{(1+t) \frac{d}{dt}(2t) - 2t \frac{d}{dt}(1+t)}{(1+t)^2} \Rightarrow \frac{dy}{dt} = \frac{(1+t)(2) - 2t(1)}{(1+t)^2}$$

$$\Rightarrow \frac{dy}{dt} = \frac{2+2t-2t}{(1+t)^2} \Rightarrow \frac{dy}{dt} = \frac{2}{(1+t)^2}$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \Rightarrow \frac{dy}{dx} = \frac{\frac{2}{(1+t)^2}}{\frac{-2}{(1+t)}} \Rightarrow \frac{dy}{dx} = \frac{2}{(1+t)^2} \times \frac{(1+t)}{-2} \Rightarrow \frac{dy}{dx} = -\frac{1}{(1+t)}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{dy}{dx} \left[-\frac{1}{(1+t)} \right] \Rightarrow \frac{d^2 y}{dx^2} = 0 \text{ [Ans.]}$$